

NAPAS, Russian Federation

WMO: 290230

Lat: 59.850N

Lon: 81.950E

Elev: 249

StdP: 14.56

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.2	-37.2	-47.4	0.3	-43.0	-41.6	0.5	-36.9	15.5	9.3	13.1	9.7	2.0	30	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	83.1	66.1	80.0	64.9	76.8	63.5	69.0	78.5	67.0	76.2	65.2	73.8	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.7	73.2	63.7	89.2	71.0	61.8	83.2	69.3	33.4	78.5	31.8	76.4	30.3	73.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.2	11.1	DB	-46.6	88.1	7.0	2.9	-51.6	90.2	-55.7	91.9	-59.7	93.5	-64.8	95.7	(4)
(5)			WB	-45.4	71.7	6.0	2.8	-49.7	73.7	-53.2	75.3	-56.6	76.9	-60.9	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	-6.9	0.4	16.3	30.1	45.3	60.7	64.1	57.2	46.5	31.8	10.0	-2.9		
	DBStd	27.46	16.65	14.39	12.15	11.14	11.03	8.37	6.65	6.66	8.09	9.91	16.90	17.38		
	HDD50	8620	1763	1388	1046	598	228	13	0	13	162	568	1200	1641		
	HDD65	13084	2228	1808	1511	1047	616	177	99	254	557	1031	1650	2106		
	CDD50	1152	0	0	0	1	83	335	438	237	56	2	0	0		
	CDD65	139	0	0	0	0	5	49	72	13	0	0	0	0		
	CDH74	1288	0	0	0	0	79	462	610	130	7	0	0	0		
	CDH80	264	0	0	0	0	16	96	131	21	0	0	0	0		
	WSAvg	5.0	4.5	4.9	5.8	6.0	6.1	4.8	4.1	4.3	4.5	5.4	5.1	4.8		
	PrecAvg	23.9	1.1	0.9	1.1	1.3	2.3	2.8	3.1	3.2	2.4	2.1	2.0	1.6		
	PrecMax	31.2	2.4	2.0	2.3	3.0	5.2	6.5	5.8	6.0	5.2	3.9	3.1	3.3		
(15) Precipitation	PrecMin	16.6	0.2	0.1	0.2	0.1	0.7	0.1	0.8	0.6	0.2	0.9	1.1	0.6		
	PrecStd	3.7	0.6	0.4	0.5	0.6	1.1	1.3	1.4	1.5	1.1	0.7	0.6	0.6		
	DB	31.4	33.0	45.0	61.2	82.3	86.3	88.2	83.3	74.5	58.5	38.8	32.3			
	MCWB	30.8	31.3	36.7	47.9	61.0	65.4	69.8	67.8	61.9	50.2	36.7	31.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	26.5	28.9	39.7	54.7	74.6	82.9	83.4	77.8	69.1	52.2	35.1	28.5			
	2% MCWB	25.6	27.4	33.7	44.6	57.7	64.7	67.8	64.6	58.4	47.1	34.0	27.3			
	5% DB	21.0	24.6	35.9	50.0	68.9	79.6	80.2	73.7	64.3	48.6	33.3	24.8			
	5% MCWB	19.8	23.1	31.6	42.0	55.0	63.7	66.3	63.4	56.0	44.8	32.3	23.8			
	10% DB	16.2	20.1	32.8	45.6	63.0	76.0	77.2	69.5	59.9	44.7	31.3	21.1			
	10% MCWB	15.2	18.6	29.8	38.9	52.4	62.5	64.7	61.1	53.8	42.1	30.3	20.2			
	0.4% WB	30.6	31.6	37.5	49.9	62.7	70.2	73.5	69.5	63.6	51.2	36.9	31.6			
	0.4% MCDB	31.0	32.6	43.8	57.7	77.1	80.1	83.7	78.8	72.0	55.9	38.0	32.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	25.5	27.7	34.3	45.7	59.1	67.3	70.1	66.5	59.5	48.0	34.1	27.5			
	2% MCDB	26.4	29.1	38.1	53.5	72.7	77.3	79.3	75.0	67.1	51.8	34.7	28.4			
	5% WB	20.1	23.1	32.2	42.4	56.3	65.6	68.1	64.3	56.8	45.2	32.6	23.9			
	5% MCDB	21.1	24.7	35.3	48.9	66.6	75.4	77.5	71.6	63.2	48.2	33.4	24.7			
	10% WB	15.2	18.8	29.9	39.4	53.1	63.8	66.3	62.0	54.1	41.9	30.3	20.1			
	10% MCDB	16.2	20.1	33.0	45.1	62.6	73.8	74.8	68.2	58.8	44.4	31.3	21.1			
	MDBR	14.6	17.1	19.3	18.2	17.3	17.7	17.3	16.0	14.4	10.4	12.4	13.9			
	5% DB	14.8	13.8	19.0	22.6	25.7	24.0	22.6	22.9	23.3	16.0	9.3	16.1			
(36) Mean Daily Temperature Range	MCWBR	14.7	13.2	14.5	14.3	12.5	9.5	9.5	11.5	12.9	10.9	9.0	15.3			
	MCDBR	14.7	13.8	17.4	20.1	23.3	20.6	20.5	20.1	20.7	14.7	9.1	15.9			
	MCWBR	14.7	13.4	13.9	13.7	12.2	9.6	9.8	11.3	12.3	10.6	8.8	15.5			
	MCDBR	14.7	13.4	13.9	13.7	12.2	9.6	9.8	11.3	12.3	10.6	8.8	15.5			
(40) Clear-Sky Solar Irradiance	taub	0.268	0.304	0.353	0.375	0.387	0.403	0.447	0.411	0.361	0.329	0.278	0.261			
	taud	1.970	2.003	1.951	2.053	2.278	2.322	2.172	2.264	2.363	2.263	2.023	1.913			
	Ebn at Noon	147	209	235	257	266	263	246	246	240	205	141	98			
	Edh at Noon	16	28	40	43	37	37	42	35	27	21	15	11			
(44) All-Sky Solar Radiation	RadAvg	155	451	934	1418	1556	1788	1616	1183	733	342	153	83			
	RadStd	15	36	60	140	179	177	170	144	96	36	33	8			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page